Data Cleaning and Preparation

Slides for *MATH 251 – Database Queries*

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Outline

1. Exploratory Data Analysis (EDA)
   - Univariate Analysis
   - Multivariate Analysis

Exploratory Data Analysis (EDA)

- The goal of **EDA** is to **understand your dataset before analysis**:
  - Identify the **data type**:
    structured, semi-structured, or unstructured
  - If structured, find the **schema**:
    Which **attributes (columns)** describe the data?
  - For each attribute, understand its **data type**:
    - Categorical (names, labels)
    - Ordinal (ordered categories)
    - Numerical (numbers)
  - Explore the **values**:
    - Typical value (average)
    - Range (min $\rightarrow$ max)
    - General pattern of the data

EDA (Cont.)

- For **structured data**, EDA has **two main steps**:
  - **Step 1: Univariate Analysis**
    Study **one attribute at a time**
  - First, identify the **domain type**:
    - Categorical → strings (e.g., names)
    - Numerical → numbers
    - Temporal → dates/times
    - Ordinal → strings or numbers (depends)
  - **Important:**
    Data types can be **misleading** → always **verify with EDA**
  - **Step 2: Multivariate Analysis**
    Study **relationships between attributes**

EDA: Univariate Analysis

- Once we know the **domain type**, we explore **its values**:
  - **Categorical data**:
    Use histograms / bar charts
  - **Numerical data**:
    Focus on:
    - Center → mean, median, mode
    - Spread → std, quartiles
  - **Optional:**
    Fit a distribution (e.g., normal) to summarize data
- **Also check for problems:**
  - Missing values
  - Outliers
  - Errors (wrong type, wrong values)
- These checks help with **data cleaning**

EDA: Univariate Analysis — Numerical Attributes

- We analyze **numerical data** using:
  - **Range**: minimum, maximum, range
  - **Center (central tendency)**: mean, median, mode
  - **Spread (dispersion)**: standard deviation, variance
  - **Shape (optional)**: skewness, kurtosis
- **Example (SQL summary): Attr is the attribute**

```
SELECT COUNT(Attr) AS number_values,
       COUNT(DISTINCT Attr) AS cardinality,
       MIN(Attr) AS minimum,
       MAX(Attr) AS maximum,
       MAX(Attr) - MIN(Attr) AS range,
       AVG(Attr) AS mean,
       STDDEV(Attr) AS standard_deviation
FROM Data;
```

Histograms: Frequency of Different Data Values

What is a Histogram?

- A histogram shows how data is **distributed**
- It groups values into **bins (ranges)**
- **Frequency**: each bar shows how many values fall in that range

Example: Exam Scores

Range	# Students
0–50	2
50–60	5
60–70	8
70–80	12
80–90	7
90–100	3

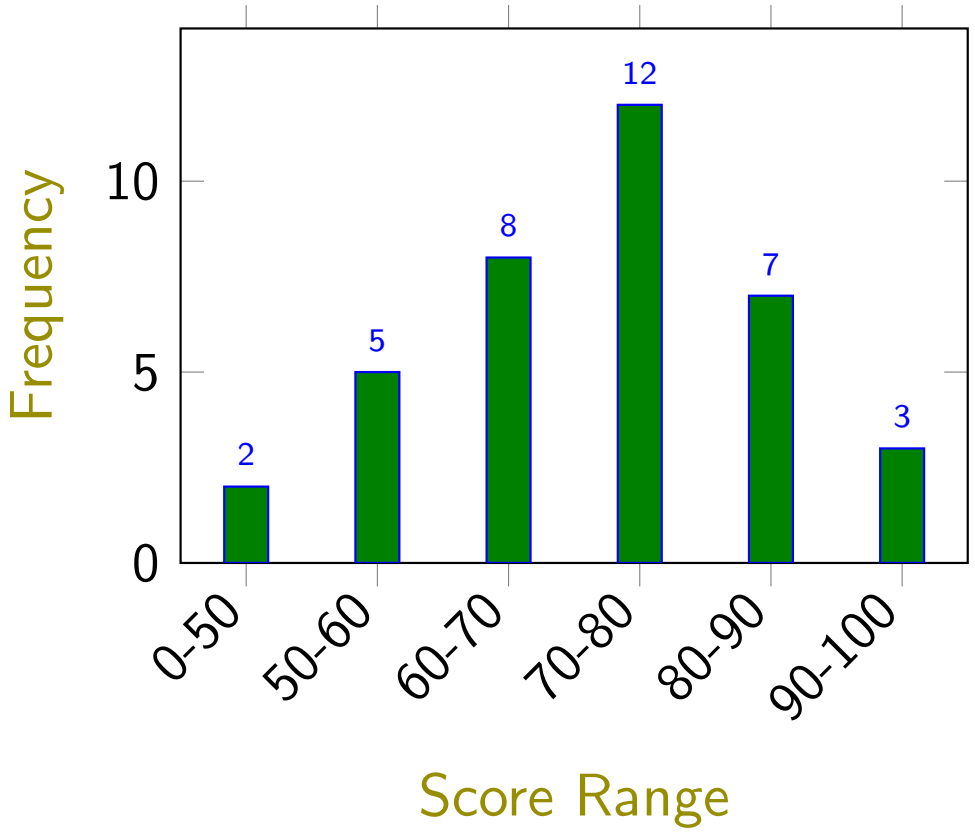
How to Read It

- **Taller bar** = more data in that range
- The **shape** shows the pattern:
  - **Symmetric** → balanced data
  - **Right-skewed** → long tail on the right
  - **Left-skewed** → long tail on the left

Key Idea: Histogram = "Where are most of the data values?"

Histogram: Visual Example

Exam Scores Distribution



How to Read It

- Each **bar** = count of data in a range
- **Taller bar** ⇒ more data
- Data grouped into **bins**

Pattern (Shape)

- **Symmetric** → balanced
- **Right-skewed** → tail on right
- **Left-skewed** → tail on left

Key Idea: Histogram = **"Where is most of the data?"**

Median: Middle/center of Ordered Data

Median

Odd Number of Values

2 4 **7** 9 12

Median = 7

Even Number of Values

2 4 **6** **8** 9 12

Median = (6 + 8)/2 = 7

How to Find It

- The **median** is the middle value
- First put the data in **ascending order**
- Then check how many values there are

Cases

- **Odd number** of values → one middle value
- **Even number** of values → average of the two middle values

Challenge

- For large datasets, sorting can take time
- In SQL, finding the median is harder than finding the mean

EDA: Numerical Attributes — Mean (Histogram)

Mean from a histogram

```
WITH Histogram(Value, Frequency) AS (
    SELECT Attr, COUNT(*)
    FROM Data
    GROUP BY Attr
)
SELECT (1.0 * SUM(Value * Frequency)) / SUM(Frequency) AS mean
FROM Histogram;
```

- We create a **histogram table** with two columns:
  - **Value**: each distinct value of the attribute
  - **Frequency**: number of times that value appears
- A **weighted average**: multiply each value by its frequency
- Divide by total number of values → **mean**

EDA: Numerical Attributes — Mean (Probability)

Mean from probabilities

```
WITH NHistogram(Value, Prob) AS (
    SELECT Attr, SUM(1.0 / total)
    FROM Data,
         (SELECT COUNT(*) AS total FROM Data) AS Temp
    GROUP BY Attr
)
SELECT SUM(Value * Prob) AS mean
FROM NHistogram;
```

- We create a table with:
  - **Value**: each distinct value
  - **Prob**: probability of that value (frequency ÷ total number of rows)
- This converts counts into **probabilities**
- Compute mean as: sum of (value × probability)

EDA: Mean via Probabilities (Step-by-Step)

Example dataset

$$\text{Data} = [2, 2, 3, 3, 3], \quad \text{total} = 5$$

Step 1: Attach total to each row (cross join)

```
Attr | total
------------
  2  |   5
  2  |   5
  3  |   5
  3  |   5
  3  |   5
```

Step 2: Compute probabilities

Each row contributes $\frac{1}{5}$, so grouping gives:

$$\mathbb{P}(2) = \frac{2}{5}, \quad \mathbb{P}(3) = \frac{3}{5}$$

Example

Trying this with dbfiddle:
`https://www.db-fiddle.com/f/pqwqGBYGDGiGNhczcJh6R5/0`

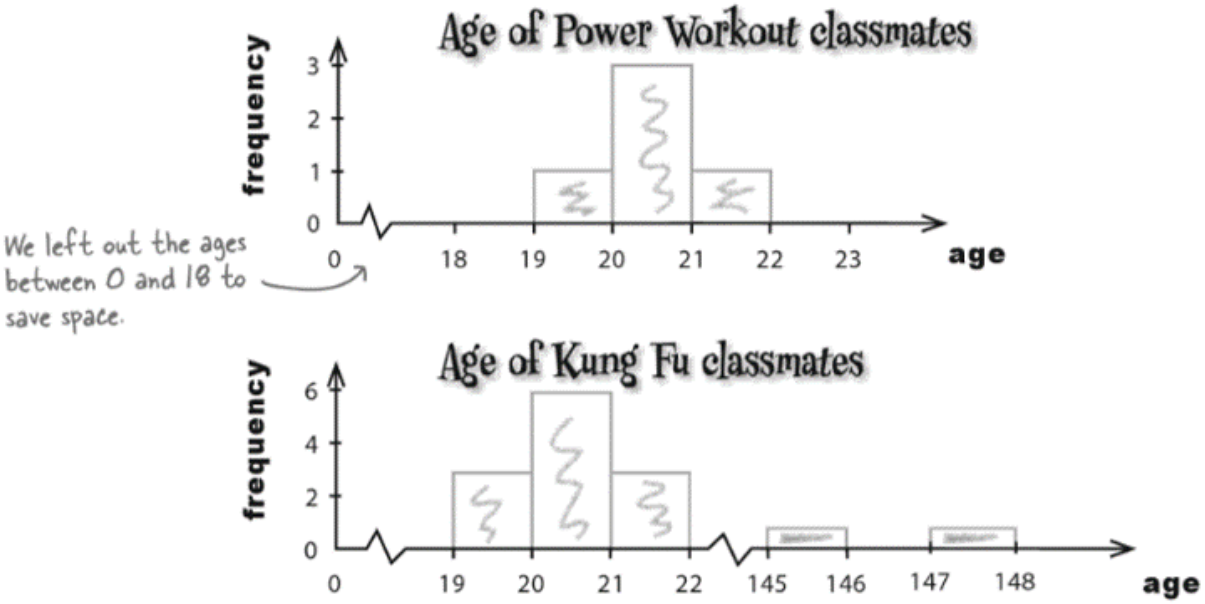
At the Health Club

Power Workout Classmate Ages

Age	19	20	21
Frequency	1	3	1

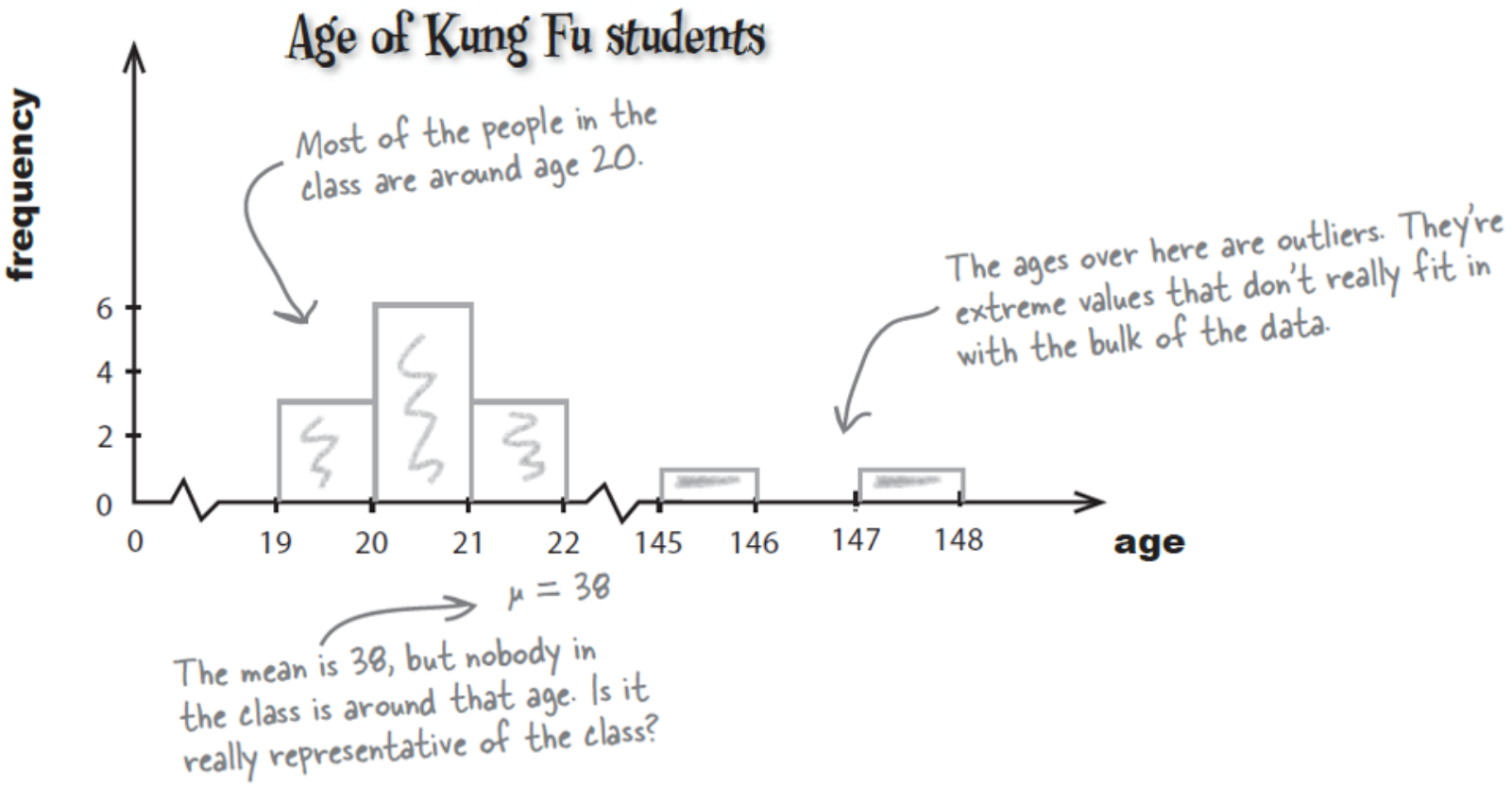
Kung Fu Classmate Ages

Age	19	20	21	145	147
Frequency	3	6	3	1	1



Source: Head First Statistics by Dawn Griffiths, O'Reilly.

At the Health Club

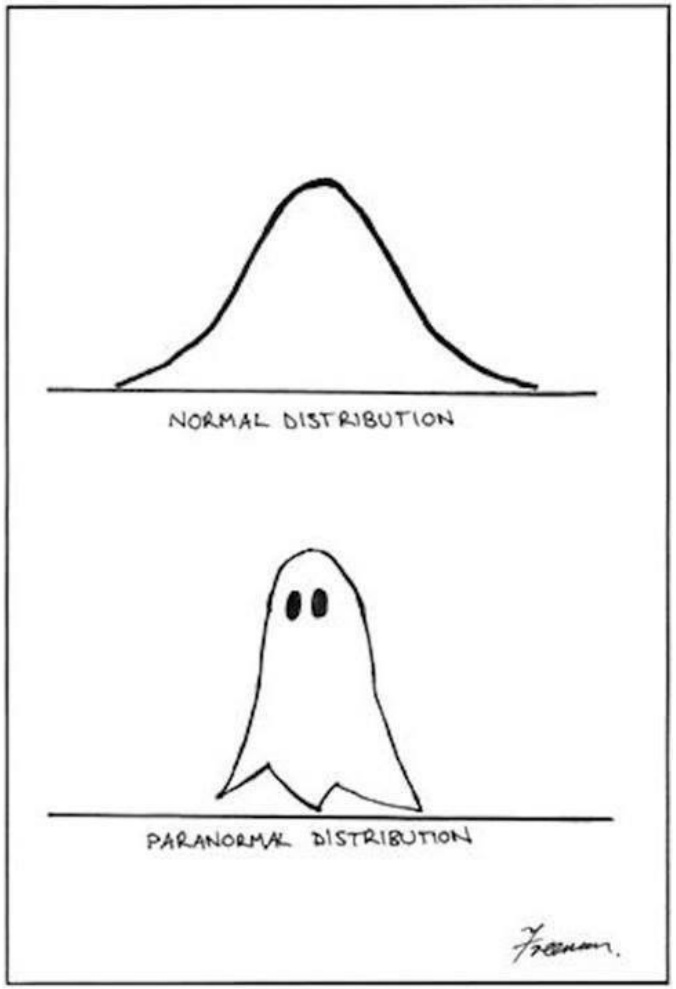


Source: Head First Statistics by Dawn Griffiths, O'Reilly.

Skewed Data

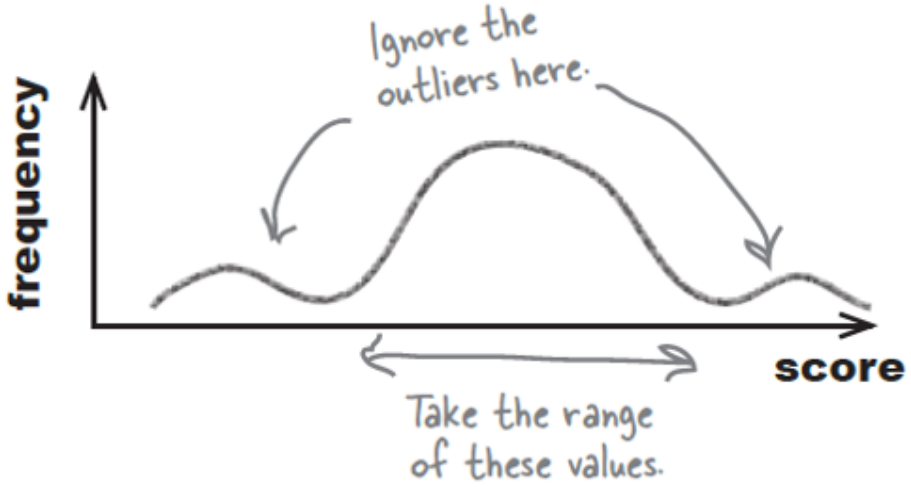
Source: Head First Statistics by Dawn Griffiths, O'Reilly.

Normal Distributions



https://www.rebresearch.com/blog/ab-normal-statistics-and-joke/.

Trimming Outliers



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EDA: Numerical Attributes — Trimmed Mean

Trimmed Mean

```
SELECT AVG(A)
FROM Data,
     (SELECT MAX(A) AS Amax FROM Data) AS T1,
     (SELECT MIN(A) AS Amin FROM Data) AS T2
WHERE A < Amax AND A > Amin;
```

Logic:

- Find the **largest** value and the **smallest** value
- Remove those **extreme values**
- Compute the average of the **remaining values**

Why do this?

It helps reduce the effect of **outliers**.

Try it in dbfiddle:

https://www.db-fiddle.com/f/442TTemmHKLqBK8u6mssHD/0

EDA: Univariate Analysis — Numerical Attributes — Mean

Geometric mean (gm) is useful when values change by **multiplication**, such as growth rates or interest rates.

- **Example:** interest rates over 5 years: 1%, 1.5%, 2%, 2.5%, 3%
- Convert them to **growth factors**: 1.01, 1.015, 1.02, 1.025, 1.03
- The **geometric mean** is $gm = (1.01 \times 1.015 \times 1.02 \times 1.025 \times 1.03)^{\frac{1}{5}}$
- If you invest **\$1000**, then after 5 years:

$$\$1000 \times 1.01 \times 1.015 \times 1.02 \times 1.025 \times 1.03 = \$1000 \times gm^5$$

- So, the geometric mean gives one **average multiplicative rate** for all 5 years.

Geometric Mean in SQL — Idea

Goal: Compute **geometric mean**

Challenge: SQL does **SUM**, not product

Key Trick: Logs

$$\ln(a \times b \times c) = \ln a + \ln b + \ln c$$

Step 1: Sum of logs

```
SELECT SUM(LN(Attr))
FROM Data;
```

Step 2: Take average

```
SELECT AVG(LN(Attr))
FROM Data;
```

Geometric Mean in SQL — Final Step

Step 3: Convert back using EXP

```
SELECT EXP(AVG(LN(Attr)))
FROM Data;
```

Final Idea:

$$gm = \exp\left(\text{avg}(\ln x)\right)$$

Interpretation:

- Log turns **product** $\rightarrow$ **sum**
- Average gives the **per-value rate**
- EXP returns to original scale

Example

Trying this with dbfiddle:
`https://www.db-fiddle.com/f/ifBmbzfDTqQptgaSUzSPVB/0`

EDA: Univariate Analysis — Numerical Attributes — Mean

Harmonic Mean is used for **rates** (like speed)
Example:

- Car goes at **60 mph** (first half)
- Returns at **30 mph** (second half)

Important: The car spends **more time** at 30 mph
Average speed is NOT 45 mph!
Correct average (Harmonic Mean): 40 mph
Key idea:

- Use harmonic mean when averaging **rates**
- It accounts for time differences

EDA: Univariate Analysis — Numerical Attributes — Mean

Harmonic Mean uses **reciprocals**

- For positive numbers $x_1, x_2, \ldots, x_n$, $\mathrm{HM} = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}}$.
- **Example:** For $x_1 = 2$, $x_2 = 3$, $\mathrm{HM} = \frac{2}{\frac{1}{2}+\frac{1}{3}} = \frac{2}{\frac{5}{6}} = \frac{12}{5} = 2.4$.

Idea:

- Take reciprocal of each value: $\frac{1}{x}$
- Compute the **average**
- Take reciprocal again

SQL Implementation:

```
SELECT COUNT(Attr) / SUM(1.0/Attr)
FROM Data;
```

Key use: Averaging **rates** (speed, price per unit, etc.)

EDA: Univariate Analysis -Numerical Attributes -Mode

Mode = most frequent value

Step 1: Build Histogram (value, frequency)

```
WITH Histogram AS (
  SELECT Attr AS val, COUNT(*) AS freq
  FROM Data
  GROUP BY Attr
)
```

Step 2: Find highest frequency

```
SELECT val
FROM Histogram,
     (SELECT MAX(freq) AS top FROM Histogram) AS T
WHERE freq = top;
```

Output: value(s) with max frequency

EDA: Univariate Analysis -Numerical Attributes -Mode

Example: How Histogram works

Data (Attr):

`10, 20, 20, 30, 30, 30`

Step 1: Histogram (value, frequency)

val	freq
10	1
20	2
30	3

Step 2: Find highest frequency

- **MAX(freq)** $= 3$
- Value with freq $= 3 \rightarrow 30$

Mode = 30

EDA: Univariate Analysis -Numerical Attributes -Mode

Simpler case: one mode

```
SELECT Attr, COUNT(*) AS freq
FROM Data
GROUP BY Attr
ORDER BY freq DESC
LIMIT 1;
```

Key idea:

- **GROUP BY** → count values
- **ORDER BY** → highest first
- **LIMIT 1** → pick top

Example

Trying this with dbfiddle:
https://www.db-fiddle.com/f/mXQC8CtV6zYYbjZARdfJZ8/0

EDA: Univariate Analysis -Numerical Attributes -Median

Median = middle value after sorting

```
SELECT AVG(Attr)
FROM (
  SELECT Attr
  FROM Data
  ORDER BY Attr
  LIMIT 2 - MOD((SELECT COUNT(*) FROM Data), 2)
  OFFSET GREATEST(CEIL((SELECT COUNT(*) FROM Data) / 2.0 - 1),0)
) AS T;
```

- **ORDER BY** → sort values
- **OFFSET** → jump to middle
- **LIMIT** → take 1 or 2 values
- **AVG** → return median

Cases:

- **Odd size** → 1 middle value (LIMIT = 1)
- **Even size** → 2 middle values → average

EDA: Univariate Analysis - Numerical Attributes - Median

```
SELECT AVG(Attr)
FROM (
  SELECT Attr
  FROM Data
  ORDER BY Attr
  LIMIT 2 - MOD((SELECT COUNT(*) FROM Data), 2)
  OFFSET GREATEST(CEIL((SELECT COUNT(*) FROM Data) / 2.0 - 1),0)
) AS T;
```

- **MOD(n,2)** → checks **odd/even** (`MOD` remainder after division)
  - odd → 1 → LIMIT = 1
  - even → 0 → LIMIT = 2
- **CEIL(n/2 - 1)** → finds **middle index** (`CEIL` rounds a number up to the nearest integer)
  - shifts index to correct starting point
- **GREATEST(...,0)** → avoids **negative offset**
  - A defensive guard when the table is empty.
- **OFFSET** → skips to middle (`OFFSET k` ignores the first k rows, then start returning results.)
- **LIMIT** → selects 1 or 2 values
- **AVG** → final median

EDA: Univariate Analysis -Numerical Attributes -Standard Deviation

Built-in function in most SQL systems:

```
SELECT STDDEV(Attr) FROM Data;
```

What it measures: how spread out the data is

Formula:

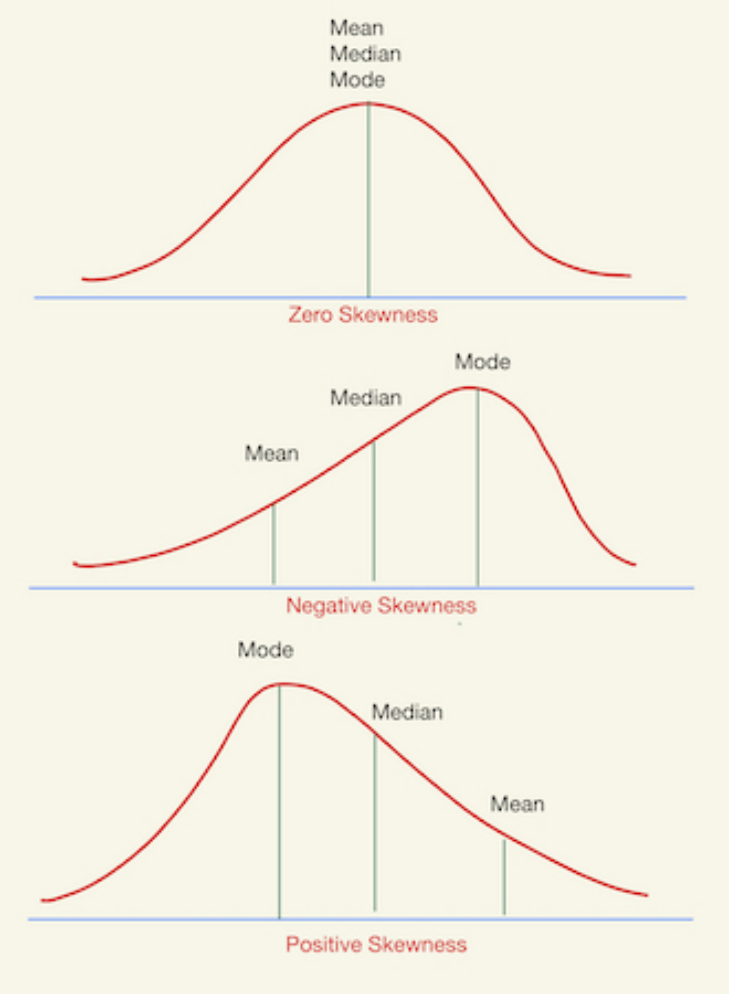
$$\sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n-1}}$$

SQL version:

```
SELECT SQRT(SUM(POWER(Attr - mean, 2)) / (COUNT(*) - 1))
FROM Data, (SELECT AVG(Attr) AS mean FROM Data) AS T;
```

Idea: distance from mean $\rightarrow$ square $\rightarrow$ average $\rightarrow$ square root

Skew



https://study.com/academy/lesson/skewness-in-statistics-definition-formula-example.html

EDA: Univariate Analysis -Numerical Attributes -Skewness

Skewness = how asymmetric the data is

Interpretation:

- **Skewness** $> 0 \rightarrow$ tail to the right
- **Skewness** $< 0 \rightarrow$ tail to the left
- **Skewness** $\approx 0 \rightarrow$ symmetric

Formula for X: data values μ: mean of the data

$$\text{Skewness} = \frac{\mathbb{E}[(X-\mu)^3]}{(\mathbb{E}[(X-\mu)^2])^{3/2}}$$

```
SELECT SUM(POW(Value - mean, 3)) / COUNT(*) /
       POW(SUM(POW(Value - mean, 2)) / (COUNT(*) - 1), 3 / 2)
FROM Data, (SELECT AVG(Attr) AS mean FROM Data) AS T;
```

EDA: Univariate Analysis -Numerical Attributes -Kurtosis

Kurtosis = how heavy the tails / how peaked the data is

Interpretation:

- **High kurtosis** → heavy (long) tails extending far out, **more outliers**
- **Low kurtosis** → light (short) tails quickly shrinking , **fewer outliers**
- **Normal** → moderate tails (baseline)

Formula for X: data values μ: mean of the data

$$\text{Kurtosis} = \frac{\mathbb{E}[(X-\mu)^4]}{(\mathbb{E}[(X-\mu)^2])^2}$$

```
SELECT SUM(POW(Value - mean, 4)) / COUNT(*) /
       POW(SUM(POW(Value - mean, 2)) / (COUNT(*) - 1), 2)
FROM Data, (SELECT AVG(Attr) AS mean FROM Data) AS T;
```

EDA: Univariate Analysis – Categorical Attributes

Build a Histogram

SQL Pattern:

```
SELECT A, COUNT(*)
FROM R
GROUP BY A;
```

- **A** = categorical attribute
- **COUNT(*)** = frequency
- Groups rows with the same value of **A**

Goal: Count how many times each category appears

Example Table: `Students`

ID	Major	Year	GPA
1	CS	Soph	3.5
2	Math	Fresh	3.8
3	CS	Junior	3.2
4	Bio	Soph	3.6
5	CS	Senior	3.4
6	Math	Soph	3.7
7	Bio	Fresh	3.3

Histogram of `Major`

Major	Count
CS	3
Math	2
Bio	2

Interpretation: CS appears most frequently

EDA: Univariate Analysis – Entropy (Idea)

Entropy measures how much **uncertainty** or **disorder** exists in an attribute.

$$\mathbf{H(A)} = -\sum_{\mathbf{a} \in \mathbf{A}} \mathbf{P(a)} \log_2 \mathbf{P(a)}$$

- $\mathbf{P(a)}$ = probability of value *a*
- If one value dominates → **low entropy**
- If values are evenly spread → **high entropy**

Think:

- **Low entropy** = predictable
- **High entropy** = uncertain / mixed

EDA: Univariate Analysis – Entropy (SQL)

How to compute entropy in SQL:

Step 1: Compute probability of each value **Step 2:** Apply entropy formula

```
SELECT -SUM(Pa * LOG(2, Pa))
FROM (
    SELECT A, COUNT(*) * 1.0 / total AS Pa
    FROM Data
    CROSS JOIN (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY A, total
) AS S;
```

What is happening?

- **GROUP BY A** → counts each value
- **1.0 / total** → builds probabilities
- **SUM(...)** → applies entropy formula

Try it: `https://www.db-fiddle.com/f/bKbjDc3ZewvgXFy4czvRmx/0`

EDA: Multivariate Analysis

Idea: Understand relationships between attributes

Depends on attribute types:

- **Categorical – Categorical**
  - Contingency tables
  - Chi-square test
- **Categorical – Numerical**
  - ANOVA (analysis of variance) checks whether different groups have significantly different averages.
  - Logistic regression
- **Numerical – Numerical**
  - Correlation, covariance
  - Linear regression
- **Ordinal – Ordinal**
  - Spearman's rank
  - Kendall's rank

EDA: Multivariate Analysis

Two ways to think about relationships

- **1. Cause & Effect (Dependence)**
  - One variable **influences** the other
  - **Independent (Predictor)** $\rightarrow$ input
  - **Dependent (Outcome)** $\rightarrow$ result
- **2. No Direct Dependence**
  - Variables may still be **related**
  - A third variable Z can affect both
  - This creates a **hidden connection**

Key Idea: Correlation does NOT imply causation

EDA: Multivariate Analysis - Joint Probability

Goal: Measure relationship between two attributes A and B
Key probabilities (SQL idea):

$$P(A) = \frac{\#A}{N}, \quad P(B) = \frac{\#B}{N}, \quad P(A, B) = \frac{\#(A, B)}{N}$$

SQL (Joint Probability):

```
SELECT A, B, SUM(1.0/total) AS JointProb
FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
GROUP BY A, B;
```

Idea: Create a table with $P(A)$, $P(B)$, $P(A, B)$
Independence Test: $P(A, B) = P(A)\,P(B)$
If true $\Rightarrow$ A and B are **independent**

EDA: Joint Probability - Example

Example: $A = \text{Gender}, \quad B = \text{Pass Exam}$

Data: **Total–** $N = 100$

Gender	Pass	Count
Male	Yes	30
Male	No	10
Female	Yes	20
Female	No	40

Probabilities: $P(\text{Male}) = \frac{40}{100}, \quad P(\text{Yes}) = \frac{50}{100}, \quad P(\text{Male, Yes}) = \frac{30}{100}$

Independence Check: $P(\text{Male, Yes}) \stackrel{?}{=} P(\text{Male}) \times P(\text{Yes})$

$$0.30 \neq 0.4 \times 0.5 = 0.20$$

Conclusion: **Not independent** $\Rightarrow$ there is a relationship

EDA: Multivariate Analysis - Independence Check

Goal: Check whether two attributes A and B are independent

Idea: If A and B are independent, then $P(A, B) = P(A)\,P(B)$ for all pairs (all combinations) of A and B values.

SQL idea:

- Compute $P(A)$, $P(B)$, $P(A, B)$
- Add up all differences (all combinations of A and B values):

$$|P(A, B) - P(A)P(B)|$$

If the final sum is 0: A **and** B **are independent**

Independence Check - Example (All Differences)

Example: $A = \text{Gender}, \quad B = \text{Pass}$

Probabilities:

$$P(\text{Male}) = 0.4,\ P(\text{Female}) = 0.6,\ P(\text{Yes}) = 0.5,\ P(\text{No}) = 0.5$$

Compute for each pair (A, B)**:** $|P(A, B) - P(A)P(B)|$

Pair	Calculation	Value
(Male, Yes)	$\lvert 0.30 - (0.4 \times 0.5)\rvert$	0.10
(Male, No)	$\lvert 0.10 - (0.4 \times 0.5)\rvert$	0.10
(Female, Yes)	$\lvert 0.20 - (0.6 \times 0.5)\rvert$	0.10
(Female, No)	$\lvert 0.40 - (0.6 \times 0.5)\rvert$	0.10

Sum of all differences: $0.10 + 0.10 + 0.10 + 0.10 = 0.40$

Conclusion: Sum $\neq 0 \Rightarrow$ **not independent**

EDA: Multivariate Analysis - Independence Check

```
WITH ProbA AS
   (SELECT A, SUM(1.0/total) AS PrA
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY A),
  ProbB AS
   (SELECT B, SUM(1.0/total) AS PrB
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY B),
  ProbAB AS
   (SELECT A, B, SUM(1.0/total) AS PrAB
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY A, B)
SELECT SUM(ABS(PrAB - (PrA * PrB)))
FROM (SELECT ProbA.A, ProbB.B, PrA, PrB, PrAB
      FROM ProbA, ProbB, ProbAB
      WHERE ProbA.A = ProbAB.A AND ProbB.B = ProbAB.B) AS Probabilities;
```

EDA: Multivariate Analysis - PMI

Pointwise Mutual Information (PMI)

Idea: How surprising is it to see A and B together?

$$PMI(A, B) = \log_2 \frac{P(A, B)}{P(A)\,P(B)}$$

Interpretation:

- **PMI = 0** $\Rightarrow$ independent
- **PMI > 0** $\Rightarrow$ occur together **more than expected**
- **PMI < 0** $\Rightarrow$ occur together **less than expected**

SQL:

```
SELECT LOG(2, PrAB / (PrA * PrB))
FROM Probabilities;
```

EDA: Multivariate Analysis - PMI

```
WITH ProbA AS (
    SELECT A, SUM(1.0 / total) AS PrA
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY A),
ProbB AS (
    SELECT B, SUM(1.0 / total) AS PrB
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY B),
ProbAB AS (
    SELECT A, B, SUM(1.0 / total) AS PrAB
    FROM Data, (SELECT COUNT(*) AS total FROM Data) AS T
    GROUP BY A, B)
SELECT A, B,
       LOG(2, PrAB / (PrA * PrB)) AS PMI
FROM (
    SELECT ProbA.A, ProbB.B, PrA, PrB, PrAB
    FROM ProbA, ProbB, ProbAB
    WHERE ProbA.A = ProbAB.A
      AND ProbB.B = ProbAB.B) AS Probabilities;
```

EDA: Multivariate Analysis - Mutual Information

Mutual Information (MI)

Idea: Average of PMI over **all pairs** (A, B)

$$I(A, B) = \mathbb{E}_{P(A,B)}[PMI(A, B)]$$

Meaning:

- Measures **overall dependence**
- Always ≥ 0
- **0** $\Rightarrow$ independent

Intuition: PMI = one pair MI = **average over all pairs**

EDA: Numerical vs Numerical - Covariance

Goal: Measure how two numerical attributes move together

Covariance: $Cov(A, B) = \mathbb{E}\big[(A - \mathbb{E}(A))(B - \mathbb{E}(B))\big]$

Equivalent formula: $Cov(A, B) = \mathbb{E}(AB) - \mathbb{E}(A)\mathbb{E}(B)$

Interpretation:

- $> \mathbf{0} \Rightarrow$ move in **same direction**
- $< \mathbf{0} \Rightarrow$ move in **opposite direction**
- $\approx 0 \Rightarrow$ weak/no linear relation

SQL:

- Built-in:

```
SELECT covar_samp(A,B) FROM Data;
```

- Manual:

```
SELECT AVG(A*B) - (AVG(A)*AVG(B))
FROM Data;
```

EDA: Numerical vs Numerical - Correlation

Pearson Correlation is a normalized covariance (scale-free)

$$\rho = \frac{Cov(A, B)}{std(A)\, std(B)}, \quad \textbf{range} : -1 \leq \rho \leq +1$$

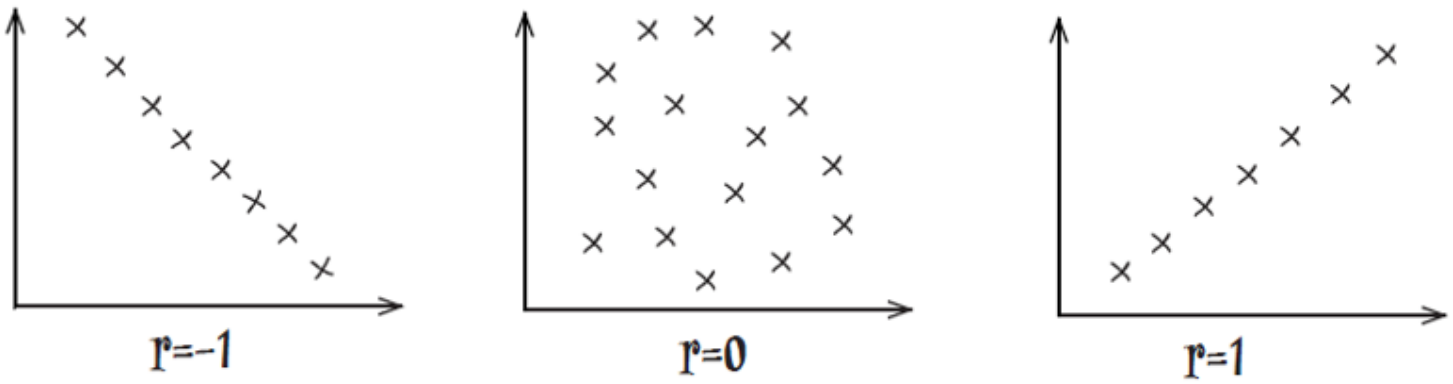
Interpretation:

- $\rho > 0 \Rightarrow$ move **together**
- $\rho < 0 \Rightarrow$ move **opposite**
- $|\rho|$ **large** $\Rightarrow$ strong relation
- $\rho \approx 0 \Rightarrow$ weak linear relation

Important:

- **If independent** $\Rightarrow \rho = 0$
- **If** $\rho = 0 \Rightarrow$ **no linear relationship**
- But there may still be a nonlinear relationship

Linear Relationships



From: "Heads First Data Statistics" by Dawn Griffiths, O'Reilly.

EDA: Numerical vs Numerical - Correlation (SQL)

Compute Correlation in SQL

Built-in (PostgreSQL):

```
SELECT CORR(A,B) FROM Data;
```

Manual (from definition):

```
SELECT (AVG(A*B) - AVG(A)*AVG(B))
       / (STD(A) * STD(B))
FROM Data;
```

Using covariance:

```
SELECT COVAR(A,B) / (STD(A) * STD(B))
FROM Data;
```